Practical 2 - Apply the what – if Analysis for data visualization. Design and generate necessary reports.

What-if-Analysis Tools

- By using What-If Analysis tools in Excel, you can use several different sets of values in one or more formulas to explore all the various results.
- For example, you can do What-If Analysis to build two budgets that each assumes a certain level of revenue.
- Or, you can specify a result that you want a formula to produce, and then determine what sets of values will produce that result.
- Excel provides several different tools to help you perform the type of analysis that fits your needs.

What-if-Analysis Tools

- Three kinds of What-If Analysis tools come with Excel: Scenarios, Goal Seek, and Data Tables.
- Scenarios and Data tables take sets of input values and determine possible results.
- A Data Table works with only one or two variables, but it can accept many different values for those variables.
- A Scenario can have multiple variables, but it can only accommodate up to 32 values.
- Goal Seek works differently from Scenarios and Data Tables in that it takes a result and determines possible input values that produce that result.

What-if-Analysis Tools

- In addition to these three tools, you can install add-ins that help you perform What-If Analysis, such as the Solver add-in.
- The Solver add-in is similar to Goal Seek, but it can accommodate more variables.
- You can also create forecasts by using the fill handle and various commands that are built into Excel.
- For more advanced models, you can use the Analysis ToolPak add-in.

Solver

- Definitions of solver - a thinker who focuses on the problem as stated and tries to synthesize information and knowledge to achieve a solution.
- Solver is a What-If analysis tool to find the optimum value in a target cells by changing values in cells used to calculate the target cells
- It is used to determine two or more unknown values
- Goal seek is used to find only one unknown value to be found
- It is an Add-in and is found in the Data Tab

- A) Problem :
- 1. Create a worksheet for a loan amount of 3,00,000, Rate of interest 6% and period is 30years. Find the loan amount that produces a monthly payment of $1500 using Goal Seek Tool.

- A) Problem :
- 2. Create a worksheet for Key Assumptions and Target variables as follows:

Input Variables (key Assumptions)	
Revenue	500000
COGS	300000
Operating Expenses	100000
Marketing Expenses	50000
R&D Expenses	20000
Interest Expenses	10000
Tax Rate	0.25
Output (Target) Variables	
Gross Profit	200000
Operating Profit	30000
Pre-Tax Profit	20000
Net-Profit	15000

- Using Scenario Manager Tool create following scenario and observe the impact of each scenario on Target Variables.

Best Case scenario:	
Revenue	600,000
COGS	280,000
Operating Expenses	90,000
Marketing Expenses	40,000
R&D Expenses	15,000
Interest Expenses	8,000
Tax Rate	0.25

Worst Case scenario:	
Revenue	400,000
COGS	320,000
Operating Expenses	110,000
Marketing Expenses	60,000
R&D Expenses	25,000
Interest Expenses	12,000
Tax Rate	0.25

B) Problem :

1. A firm makes two types of furniture's, chairs and tables. Profits are ₹ 20 per chair and ₹ 30 per table. Both products are processed on three machines M1, M2 and M3. The time required for each product in hours and total time available in hours per week on each machine are as follows:

Machine	Chair	Table	Available Time
M1	3	3	36
M2	5	2	50
M3	2	6	60

How should the manufacturer schedule his production in order to maximize profit?

- Problem :

2. Let be a company has three factories at $F1$, $F2$ and $F3$. $These$ $factries$ supply warehouses at $W1$, $W2$ and $W3$. But weekly each factory capacities are 200 units, 160units and 90 units respectively. Whereas weekly warehouses requirements are 180units, 120units and 150units respectively. It is given in tabular form unit shipping costs are.

Shipping Cost Per Unit	Warehouse		
Factory	W1	W2	W3
F1	16	20	12
F2	14	08	18
F3	26	24	16

Find the optimum distribution for this company to optimize shipping costs.